Ankita Das

Curriculum Vitae

SRM University AP, Amaravati
Mangalagiri-Mandal, Neeru Konda
Andhra Pradesh, India
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Research interest

My research interest widely lies in algebraic number theory. Currently I am interested in Jacobian varieties of super elliptic curves. Also I like to explore different aspects of algebraic geometry and the interplay between geometry and number theory.

Academic information

2024-Present **PhD in Mathematics**, *SRM University, AP*, and Indian Institute of Science Education and Research(IISER),Thiruvananthapuram

2020-2022 **Master of Science (M.Sc.) in Mathematics**, *Jadavpur University*

2017-2020 **Bachelor of Science (B.Sc.) in Mathematics**, *Jadavpur University*

2017 **Higher Secondary Examination (10+2 level), West Bengal Board**, *Sunity Academy*
Percentage – 93.6

2015 **Secondary Examination (10 level), West Bengal Board**, *Sunity Academy*
Percentage – 95.6

Papers published

2026 **Generalized Fruit Diophantine Equation And Super Elliptic Curves**, *Published in Monatshefte Fur Mathematik*, `https://doi.org/10.1007/s00605-026-02201-6`, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty

Preprints

2026 **Rational Torsion and Ramification Constraints on Odd Superelliptic Jacobians**, *arXiv preprint*, `https://arxiv.org/abs/2611.06193`, arXiv:2611.06193

Projects

Ongoing work in PhD **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty

Master's Project **A Survey on Clean Ring and Its Structural Behaviour**, link: `https://drive.google.com/file/d/1QEc2nNCNo7x_4BlHTCkSsJS3VSQXVSbH/view?usp=drivesdk`, under supervision of **Prof Sukhendu Kar**

Reading Project **Invariant Theory of $n \times n$ matrices**, *under supervision of* ***Dr. Sarbeswar Pal***

Some slides of my talks

- **On Jacobian on a Family of Superelliptic Curves**, link: `https://drive.google.com/file/d/1lvwSfrl7jE-k-ZNXUjwckJ3xArooIjCm/view?usp=drive_link`
- **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, link: `https://drive.google.com/file/d/1nq56pQEbzHCpuFzFdTPHHOibFv4wzvHf/view?usp=drivesdk`
- **From Fruits to Class Groups**, link: `https://drive.google.com/file/d/16ayXCVwTSfUipIeFV88a9r4S6PWTVU_Q/view?usp=drivesdk`
- **Factorization of Sine and Cosine functions and Proof of Walli's Formula**, link: `https://drive.google.com/file/d/1EK-sXOJOREoSGXfnHoqfGao__CXsdR4g/view?usp=drive_link`
- **Zariski-Nagata Theorem: A Few Consequences**, link: `https://drive.google.com/file/d/1mKtPXQp_n2EJIdaNx6fh-ftqd4r_NaHI/view?usp=sharing`

Courses Taken in PhD Coursework

- Fourier Analysis
- Advanced Complex Analysis
- Advanced Measure Theory
- Commutative Algebra
- Differential Geometry
- Manifold
- Algebraic Number Theory
- Algebraic Geometry
- Category Theory
- Representation Theory
- Infinite Galois Extension
- Cohomological Algebra
- Research Methodology
- Research Publication and Ethics
- Research Seminar

Courses Taken in Graduate Studies

- Abstract Algebra
- Real Analysis
- Topology
- Linear Algebra
- General Mechanics
- Differential Geometry,Probability and Stochastic Process
- Linear Algebra and Module Theory
- Complex Analysis and PDE
- Functional Analysis
- Ordinary Differential Equations and Special functions
- Advanced Topology
- Dynamical System
- Integral Equations, Integral Transforms and Calculus of Variations
- Algebraic and Differential Topology
- Advanced Rings and Modules-I
- Theory of Semigroups-I
- Optimization and Control Theory
- Advanced Functional Analysis
- Advanced Rings and Modules-II
- Theory of Semigroups-II
- Topics in Analysis
- Topics in Linear Algebra
- Topics in Manifold
- Commutative Algebra
- Topics in Algebra
- Differential Geometry

Courses Taken in Undergraduate Studies

- Algebra, Geometry and Calculus
- Abstract Algebra
- Statistics
- Real Analysis
- Ordinary Differential Equations
- Computer Science
- Linear Algebra
- Numerical Methods
- Partial Differential Equations
- Metric Space and Complex Analysis
- Linear Programming and Game Theory
- Probability
- Mechanics
- Number Theory
- Combinatorics and Graph Theory

Exams Qualified

2023 **Qualified Joint CSIR-UGC NET**

2023 **Qualified West Bengal State Level Eligibility Test**, *(WBSET)*

2021 **Qualified IIT JAM examination**

2017 **Obtained 94% marks in Mathematics, 90% marks in Physics in 10+2 standard.**

2015 **Obtained 100% marks in both Mathematics and Physical Science in 10 standard.**

Awards and Scholarships

2025 **President Scholastic Award**, *as **"Maintenance Allowance"** for securing **external government funding***, by SRM University, AP

2023-Present **Joint CSIR-UGC JRF Fellowship**, *for qualifing all India **Joint CSIR-UGC NET***, by UGC

2022-2021 **Indira Gandhi Single Girl Child Scholarship**, *by UGC*

2022-2017 **Bigyani Kanya Medha Britti (BKMB) scholarship**, *for securing position in **top 50 female candidates** in a state level exam organized by **JBNSTS***

2017 **DST-INSPIRE Scholarship**, *for securing position in **top 1 percent candidates** in 10+2 level state board examination*, by **Scholarship for Higher Education (SHE)**

Work Experience

Odd Semester 2026-2027 **TA for grading**, *CSE 203 (Discrete Mathematics)*, SRM University, AP

Odd Semester 2025-2026 **TA for grading**, *CSE 203 (Discrete Mathematics)*, SRM University, AP

2022 **Gramin Dak Sevak, Assistant Branch Post Manager**, *Indian Sub Post Office*, Nilkuthi, Cooch Behar

Technical Skills

Programming Languages **LaTex, C, C++, Python**

Conferences and Symposiums

July, 2026 **41st Annual Conference of Ramanujan Mathematical Society (RMS)**, *India*

December, 2025 **Indian Women and Mathematics (IWM), Research in Groups programme**, *India*

December, 2025 **Sakura Science Exchange Program**, *Japan*

October, 2025 **International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025**, *SRM University, AP*, Andhra Pradesh, India

January, 2025 **Frontier Symposium in Mathematics (FSM)-2025**, *IISER Thiruvananthapuram*, Kerala, India

October, 2023 **International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025**, *Kerala School of Mathematics (KSoM)*, Kerala, India

Extra Curricular Activities

- I was trained in **Hindustani Classical Music** and **Rabindrasangeet** for several years. I like to listen various forms of semi-classical and light music. I enjoy different classical instruments. I won several district and state level music competitions.
- I was trained in **recitation** from Bangiya Sangeet Parishad and also completed my **Visharad** in it. I won various state and national level recitation competitions. I was in **AIR** as child recitation artist.
- I won a national essay writing competition on **"Eradication of Poverty"** organised by **TATA building India** and felicitated by **President of India** in 2014.
- Writing poetry, stories and reading books give me immense pleasure.
- I always enjoy to take part in instantaneous debate and won many district level competitions.

Languages

English, Hindi, Bengali